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- Given the three integers a & b. Return the nth magical number. Since ans is very large return $1e^9+7$

A positive no. is magical when x is divisible by a or b

```
# include < bits/stdc++.h >
using namespace std;
#define mod 1e9+7;
int solve ( int x, int a, int b ) {
    long long low = 1;
    long long high = 1LL x n x min (a, b)
    long long l = (a x b) / -- gcd (a, b);
    while ( low < high ) {
  Long long mid = low + (high - low) / 2 ;
    int cnt = mid/a + mid/b - mid/l;
      if (cnt < n) {
          low = mid + 1;
          }
      else {
          high = mid;
      }
    }
    return low % mod ;
```

```
int main() {
    int n,a,b;
    cin>>n>>a>>b;
    cout<<solve(n,a,b);

    return 0;
}
```